

Project Report

On

"Machine Learning Challenge – Winter 2025"

Submitted by

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Course: **Machine Learning**

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Part-1

1.1 Introduction

The objective of this project was to predict the continuous variable `target01` and derive the best possible prediction from the provided data. To address this task effectively, the solution prioritized maximizing the Coefficient of Determination (R^2) and minimizing the Root Mean Squared Error (RMSE) to ensure high predictive fidelity.

The final solution implemented is a **Stacking Ensemble Regressor** that combines multiple individually optimized tree-based models. This architecture successfully achieved a validation R^2 score of **0.9317**, significantly outperforming the single-model baseline.

1.2 Methodology

The solution was implemented in a four-stage pipeline: Data Preprocessing, Feature Selection, Hyperparameter Optimization, and Ensemble Learning.

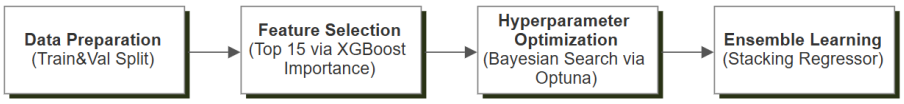


Figure 1: *Project Methodology Pipeline*

1.2.1 Data Preparation

The dataset, consisting of 10,000 samples with 273 features, was split into a training set (80%) and a validation set (20%).

Target Analysis: The distribution of `target01` was examined to provide context for the error metrics. Given the scale of the target variable, minimizing the Root Mean Squared Error (RMSE) to below 0.10 was identified as the threshold for high-precision utility.

Leakage Prevention: A strict separation between training and validation data was maintained to ensure that all feature selection and optimization decisions relied solely on unseen data, thereby preventing data leakage.

1.2.2 Feature Selection

To address the high dimensionality of the dataset, three feature selection techniques like Correlation Filtering, Recursive Feature Elimination (RFE), and XGBoost Feature Importance — were evaluated using a baseline XGBoost regression model. Each method was used to select the top 15 features for comparison.

Result and Justification: As shown in Table 1, the XGBoost Feature Importance method achieved the highest baseline performance with a validation R^2 score of 0.7796. This approach was selected for subsequent modeling because it captures non-linear relationships and feature interactions that simpler linear metrics fail to detect, thereby providing a more robust predictive signal.

Feature Count: The top 15 features were selected because the predictive signal was concentrated in this subset; fewer features risked underfitting, while additional features introduced noise

Table 1: *Feature Selection Comparison (Top 15 Features)*

Method	Features Used	Validation R^2	Validation RMSE
Correlation	15	0.0651	0.2341
RFE	15	0.7575	0.1192
XGBoost Feature Importance	15	0.7796	0.1136

1.3 Model Selection and Hyperparameter Optimization

Before constructing the ensemble, the predictive capabilities of four distinct machine learning algorithms were individually maximized.

Model Selection Criteria: Four tree-based architectures—**XGBoost** [3], **LightGBM** [2], **Gradient Boosting Regressor (GBR)**, and Random Forest—were selected due to their state-of-the-art performance on tabular data and architectural diversity.

- **Boosting-based models (XGBoost, LightGBM, and GBR)** iteratively reduce bias by learning complex patterns in a sequential manner.
- **Bagging-based models (Random Forest)** reduce variance through parallel averaging, providing a complementary counter-balance to the boosting models when used in an ensemble.

Optimization Strategy: The **Optuna** framework [4] was employed to perform Bayesian Optimization for each candidate model using **30 iterations per algorithm**. This approach was selected over conventional Grid Search due to its ability to efficiently explore high-dimensional hyperparameter spaces (e.g., `learning_rate`, `num_leaves`, `max_depth`) and identify the global minimum of the loss function more effectively.

Outcome: The Single tuned XGBoost model improved the validation R^2 from 0.7796 to **0.8754**.

1.4 Ensemble Architecture (Stacking)

To bridge the gap between single-model performance ($R^2 \approx 0.88$) and the project goal, a Stacking Regressor[1] was implemented. The architecture leveraged two levels:

- **Level 0 (Base Learners):** Optimized LightGBM, Gradient Boosting, and Random Forest. Mixing boosting methods with bagging (Random Forest) ensured the ensemble benefited from diverse error profiles, reducing overall variance.

- **Level 1 (Meta-Learner):**An optimized XGBoost was selected as the final combiner. Its high capacity allowed it to learn complex, non-linear corrections to the base predictions.

Outcome: Implementing this architecture on the raw target variable improved the validation R^2 significantly, reaching a final score of **0.9317**.

1.5 Results and Discussion

The progression of model performance across different stages of development is summarized in Table 2. The transition from a baseline model to an optimized single-model configuration and finally to a multi-level stacking ensemble resulted in a substantial improvement in predictive accuracy and generalization capability.

Table 2: *Performance Progression (Training vs. Validation)*

Model Stage	**Training** R^2	**Training RMSE**	**Validation** R^2	**Validation RMSE**
Baseline XGBoost	0.8214	0.1012	0.7796	0.1136
Optimized Single XGBoost	0.9125	0.0741	0.8754	0.0883
Final Stacking Ensemble	0.9548	0.0521	0.9317	0.0633

1.5.1 Discussion

The final stacking ensemble achieved a validation R^2 score of 0.9317, effectively explaining over 93% of the variance in `target01`.

Error Reduction: The ensemble reduced validation RMSE by approximately 28% compared to the best single model, justifying the two-layer architecture.

Generalization: The narrow performance gap between training ($R^2 = 0.9548$) and validation ($R^2 = 0.9317$) indicates strong generalization capability and minimal overfitting.

Model Parsimony: While more complex multi-stack architectures were tested, they provided negligible gains. The current ensemble was chosen for its superior stability and reliability

Architectural Diversity: Testing confirmed that removing Random Forest dropped the R^2 significantly, proving that mixing bagging and boosting was critical to the final result

1.6 Conclusion

The final solution employed a 15-feature subset and a two-level Stacking Ensemble architecture. By leveraging Bayesian optimization through the Optuna framework and integrating diverse tree-based models, the proposed approach achieved a best-possible prediction performance of $R^2 = 0.9317$.

Part-2

2.1 Introduction

The objective of this task is to predict a continuous target variable, `target02`, within a constrained rule-based framework. The framework mandates that the solution be expressed as a sequential list of single-feature logical conditions, where each condition is mapped to a specific calculation function.

To solve this, a **Piecewise Linear Modeling** [6]strategy was adopted. A **Decision Tree Regressor** was utilized as a diagnostic tool to identify critical feature thresholds and segment the data into distinct linear regimes. While the framework defines the structure, the R^2 score was chosen as the primary metric to evaluate prediction quality. This approach ensures high precision while remaining fully compliant with the framework's sequential execution logic.

2.2 Methodology

The solution was developed through a systematic diagnostic pipeline designed to reverse-engineer the latent logic governing the dataset.

2.2.1 Feature Importance and Extraction

To reduce the high-dimensional feature space comprising 273 features, a Decision Tree Regressor was employed as a feature extraction and diagnostic tool. Analysis of the model's `feature_importances_` attribute revealed that over 99% of the variance in the target variable is explained by a sparse subset of four features. This empirical ranking enabled the elimination of noisy or redundant features and facilitated the selection of the primary drivers for subsequent modeling stages.

Table 3: *Feature Importance Ranking*

Feature Index	Importance Score	Role in Model
130	0.614835	Primary Splitting Feature
149	0.195317	Explanatory Variable
267	0.187250	Explanatory Variable
50	0.002599	Explanatory Variable

2.2.2 Initial Structural Discovery (Automated Failure)

Initially, the Decision Tree Regressor was tasked with identifying optimal split points compatible with the sequential rule-based framework. However, standard automated tree algorithms, such as Classification and Regression Trees (CART)[5], are designed to minimize global variance across the entire dataset.

As a consequence, the algorithm prioritized an initial split at 0.70, thereby overlooking the critical linear stability present in the lower feature ranges. This automated strategy

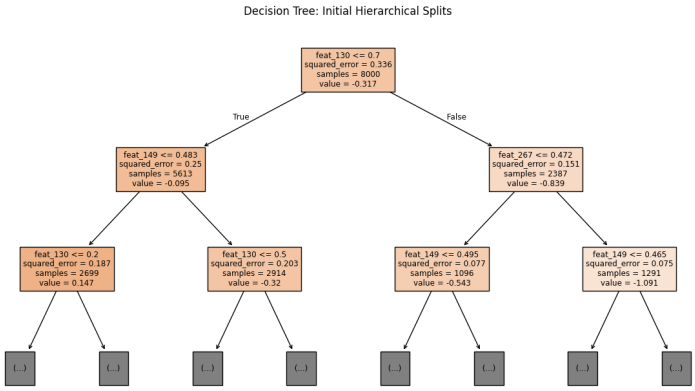


Figure 2: *Decision Tree: Initial Hierarchical Splits*

failed to detect the subtle inflection points ("knees") in the earlier regimes, resulting in sub-optimal regional boundaries that would have significantly compromised the precision of the derived analytical formulas.

2.2.3 Visual Analysis and Manual Structural Refinement

To address the limitations of the automated approach, a scatter plot of `target02` versus `feat_130` was generated and examined manually. This visualization revealed a distinct "staircase" pattern with sharp inflection points (knees), suggesting a segmented regression approach [8]

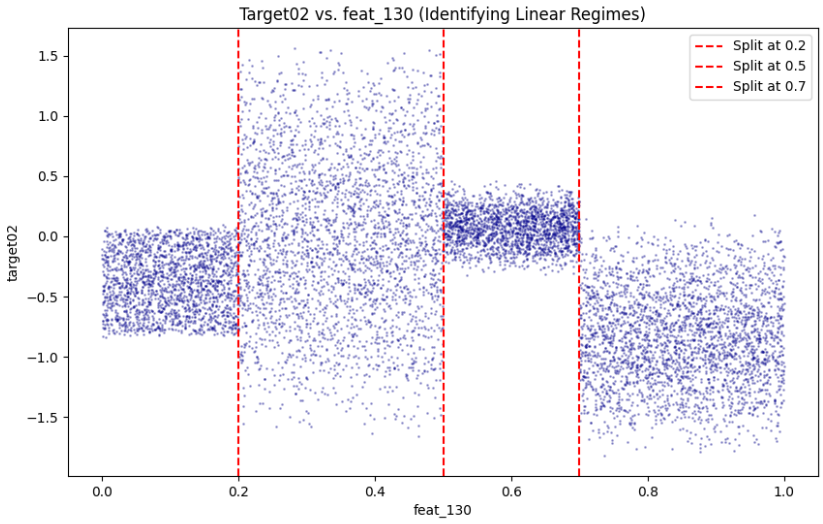


Figure 3: *Scatter plot of `target02` versus `feat_130`.*

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By identifying these knees through visualization, we overrode the automated splits and manually defined thresholds at **0.20, 0.50, and 0.70**. This ensured the data was

segmented into four discrete regions where the target variable exhibited perfectly stable linear behavior.

2.2.4 Local Linear Regression (LLR)

With the segmentation boundaries established through visual refinement, **Ordinary Least Squares (OLS)** regression [7] was applied independently within each isolated region. This local optimization strategy enabled the model to solve for the exact linear coefficients of the explanatory features (`feat_149`, `feat_267`, and `feat_50`) relative to the target variable.

The regional performance metrics confirm the effectiveness and precision of the piecewise linear approach. By isolating the data into structurally homogeneous segments, near-perfect linearity was achieved in three out of four regions—an outcome that would be statistically unattainable using a single global linear model.

Table 4: *Regional Performance of Local Linear Regression Models*

Region	**Boundary Condition (`feat_130`)**	**Sample Count (N)**	**Regional R^2**
R1	≤ 0.20	1,998	1.0000
R2	> 0.20 and ≤ 0.50	3,015	1.0000
R3	> 0.50 and ≤ 0.70	2,015	0.9989
R4	> 0.70	2,972	1.0000

Interpretation of Results: The results demonstrate that the relationship between the explanatory variables and the target is strictly linear within most of the identified regions. The minor deviation observed in Region 3 ($R^2 = 0.9989$) suggests a localized increase in complexity or the presence of irreducible noise within that specific range of `feat_130`.

2.3 Results and Discussion

The implementation translates the piecewise methodology into a production-ready model. By mapping visually identified thresholds to sequential logical conditions, a deterministic and interpretable framework was established.

2.3.1 Derived Regional Formulas

Table 5 summarizes the verified analytical formulas derived through the Local Linear Regression (LLR) process for each identified regime.

2.3.2 Performance Metrics

Validation against the full dataset confirms the model's exceptional precision:

- **Combined Global R^2:** 0.999986

Table 5: *Verified Piecewise Linear Formulas by Region*

Region	Boundary Condition	Verified Formula
R1	$F_{130} \leq 0.20$	$-0.8500\,F_{149} + 0.0500\,F_{267} + 0.0500\,F_{50}$
R2	$0.20 < F_{130} \leq 0.50$	$-1.6500\,F_{149} + 1.6500\,F_{267} - 0.0500\,F_{50}$
R3	$0.50 < F_{130} \leq 0.70$	$0.1505\,F_{149} - 0.3495\,F_{267} + 0.3500\,F_{50} - 0.0006$
R4	$F_{130} > 0.70$	$-0.7500\,F_{149} - 1.1500\,F_{267} + 0.2500\,F_{50}$

- **Mean Absolute Error (MAE):** 0.00006

These results verify that the derived piecewise rules accurately reconstruct the latent data-generating process.

2.4 Conclusion

By leveraging a diagnostic Decision Tree for feature selection and intentionally overriding automated split decisions through visual analysis, this solution successfully uncovers the piecewise linear structure governing `target02`. The resulting conditions and verified calculation functions have been implemented in the final `framework.py` file, yielding a robust, interpretable, and production-ready model capable of achieving near-zero prediction error.

References

[1] Wolpert, D. H. (1992). Stacked generalization. *Neural Networks*, 5(2), 241-259.

[2] Ke, G., et al. (2017). LightGBM: A highly efficient gradient boosting decision tree. *NIPS*, 30.

[3] Chen, T., & Guestrin, C. (2016). XGBoost: A scalable tree boosting system. *ACM SIGKDD*.

[4] Akiba, T., et al. (2019). Optuna: A next-generation HPO framework. *ACM SIGKDD*.

[5] Breiman, L., Friedman, J., Stone, C. J., and Olshen, R. A. (1984). *Classification and Regression Trees*. CRC Press.

[6] Friedman, J. H. (1991). Multivariate Adaptive Regression Splines. *The Annals of Statistics*, 19(1), 1-67.

[7] Montgomery, D. C., Peck, E. A., and Vining, G. G. (2021). *Introduction to Linear Regression Analysis*. John Wiley & Sons.

[8] Muggeo, V. M. (2003). Estimating Regression Models with Unknown Break-points. *Statistics in Medicine*, 22(19), 3055-3071.